



Parktown Boys' High School

Geography Department

Grade 11 June Exam

MEMO

Examiner: Mr. K. Clark
Mr. G. Faasen

Duration: 3 Hours

Moderator: Mr. D. Grace

Total: 150 Marks

Date: 18 June 2025

INSTRUCTIONS:

1. Write neatly and legibly
 2. Write in **BLUE PEN ONLY!!** Make sure that all questions are answered on the answer sheet provided
 3. Make sure to add your **teachers name at the top of your answer sheet**
 4. Good Luck!!
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Section A – Climatology and Geomorphology [120]

Question 1 – Multiple Choice Questions [5]

- 1) Use the words in the box below to complete the statements. Write only the correct answer, **EXAMPLE: 1.6 – River**

Global air circulation; Westerly winds; Trade winds; Geostrophic wind; Coriolis force; Pressure gradient force; Revolution; Insolation; Easterly winds; Berg winds

- 1.1 The ... causes air to be deflected from its original path. (1)
- 1.2 Large-scale movement of air in the atmosphere is referred to as ... (1)
- 1.3 The difference in air pressure causing air to move from an area of high pressure to an area of low pressure is referred to as the ... (1)
- 1.4 Winds that blows parallel to isobars in the upper atmosphere are ... (1)
- 1.5 Winds that blow in the tropics are called ... (1)

State whether the following statements are true or false [5]

- 1.6 Ocean currents are either warmer or colder than the surrounding sea. (1)
- 1.7 Cold ocean currents move water from the equator towards the poles. (1)
- 1.8 Air and water can transfer heat energy from one area to another. (1)
- 1.9 If wind blows from a tropical zone towards a cooler zone it transfers energy and cools the region. (1)
- 1.10 The Tropical Easterlies blow from subtropical regions towards tropical regions. (1)

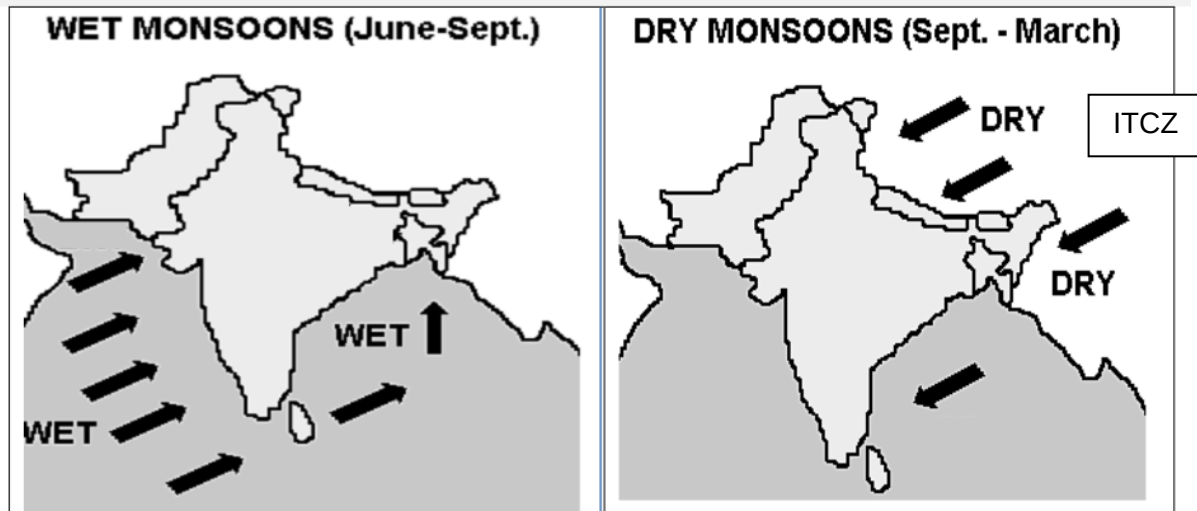
- 1.1 Coriolis force
- 1.2 Global air circulation
- 1.3 Pressure gradient force
- 1.4 Geostrophic wind
- 1.5 Trade winds
- 1.6 True
- 1.7 False
- 1.8 True
- 1.9 False
- 1.10 True

Question 2 – Monsoon [10]

Monsoon winds occur over India in the Northern hemisphere.

In the Northern Hemisphere - summer monsoons - June to September.

In the Northern Hemisphere - winter monsoons - September to March.



Look at the image below and answer the questions that follow:

2.1 **What** is a monsoon wind?

(1x2)(2)

A **monsoon wind** is a **seasonal wind pattern** that changes direction between summer and winter, typically bringing **heavy rainfall in summer** and **dry conditions in winter**.

2.2 **Identify** the wind which dominates East Africa from January to March and from June to September respectively. (2x1)(2)

From January to March: The North-East Monsoon (also called the North-East Trade Winds)

From June to September: The South-East Monsoon (also called the South-East Trade Winds)

2.3 **Describe** the weather in East Africa as a result of the wind that blows from June to September. (1x2)(2)

From **June to September**, the **South-East Monsoon** (South-East Trade Winds) blows across East Africa. These winds come from the Indian Ocean and bring **cooler, drier, and more stable conditions** to much of the region.

2.4 **State** ONE problem experienced in India associated with the monsoon wind that blows from January to March. (1x2)(2)

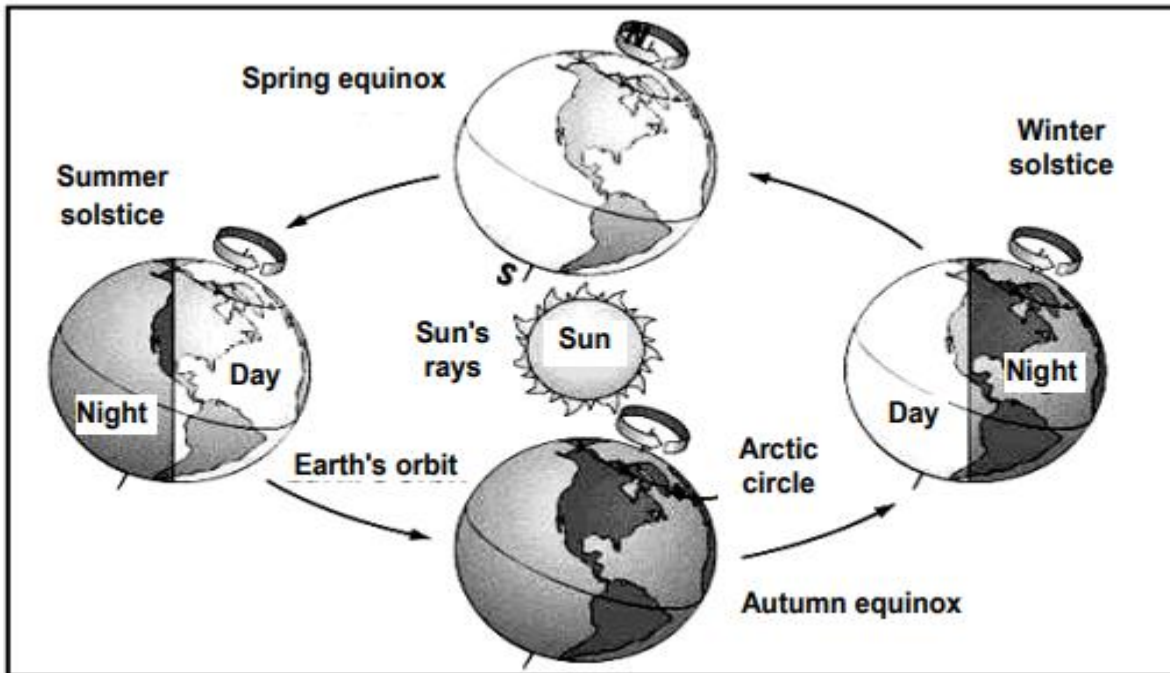
Drought and water shortages, as this wind is dry and brings very little rainfall to most parts of the country.

2.5 **State** ONE problem associated with the monsoon wind that blows from June to September. (1x2)(2)

Severe flooding, which can damage crops, homes, and infrastructure across India.

Question 3 – Earth's Energy balance [10]

Look at the Picture below, based on seasonal temperature changes, and answer the questions that follow. All information in present in the picture represents seasons for the Northern Hemisphere.



- 3.1 **What** is the name of the events that occur when the sun is directly overhead at the Equator, and day and night are equal in length? (2x1)(2)

The **equinoxes**

- 3.2 During **which** season in the Northern Hemisphere does the Arctic Circle experience 24 hours of night? (1x1)(1)

During **winter** in the Northern Hemisphere, the Arctic Circle experiences **24 hours of night** (polar night).

3.3 Refer to the Earth's position during the Winter Solstice. **Which** hemisphere is tilted away from the sun? **Explain** how this affects the amount of daylight in that hemisphere. (1+2)(3)

During the **Winter Solstice**, the **Northern Hemisphere** is tilted **away** from the sun.

Because of this tilt:

- The sun's rays hit the Northern Hemisphere at a **lower angle**, spreading energy over a larger area.
- The **days are shorter** and the **nights are longer** because the sun stays lower in the sky and rises later and sets earlier.
- This results in **less daylight hours** and colder temperatures, marking the start of winter in the Northern Hemisphere.

3.4 **Explain** the significance of the tilt of the earth for farmers in South Africa. (2x2)(4)

Tilt causes seasonal changes:

The Earth's tilt means South Africa experiences different seasons (summer and winter), which affects when farmers can plant and harvest crops **(2 marks)**

Variation in daylight and temperature:

During summer (when South Africa is tilted towards the sun), there are longer daylight hours and warmer temperatures, promoting crop growth. In winter, shorter days and cooler temperatures slow down growth, influencing farming schedules **(2 marks)**

Question 4 – El Niño and La Niña [20]

Read the following article and answer the questions that follow:

The RMS Titanic set sail on 10 April 1912 on her first voyage from Southampton to New York. The ship collided with an iceberg in the North Atlantic on 15 April 1912 and sunk, resulting in the loss of 1 522 lives. Lon S Safko, a meteorologist, recognised a direct correlation between the current warming trends experienced in the air and sea surface temperatures of the North Atlantic, and the melting of the Arctic's icebergs which break off and float southward into the cross-Atlantic shipping channels.

Research revealed that the North Atlantic experienced an El Niño event during the winter of 1911/1912 resulting in water temperatures as much as 5 °C warmer than normal. Safko believes that this could, in part, explain Captain Edward J Smith's ignorance that fateful evening, as the previous 14 years of commanding trans-Atlantic vessels had shown much dryer, colder, ice-free, non-El Niño conditions.

[Source: Historic/El Niño Information: http://www.ocregister.com/news/1997/el_nino/history/history.html]

4.1 **State** two conditions necessary for an El Niño event to occur. (2x1)(2)

Weakening or reversal of the trade winds

Warming of sea surface temperatures.

4.2 **Describe** and **explain** the relationship identified by meteorologist Lon S Safko between warming trends and iceberg movement. (2x2)(4)

Meteorologist Lon S Safko observed that as global temperatures rise (warming trends), icebergs tend to melt faster and their movement patterns change **(2 marks)**

Specifically, warming causes icebergs to break off more frequently from glaciers and drift differently due to changes in ocean currents and melting rates. This relationship shows how climate warming directly impacts iceberg behavior and the stability of polar ice **(2 marks)**

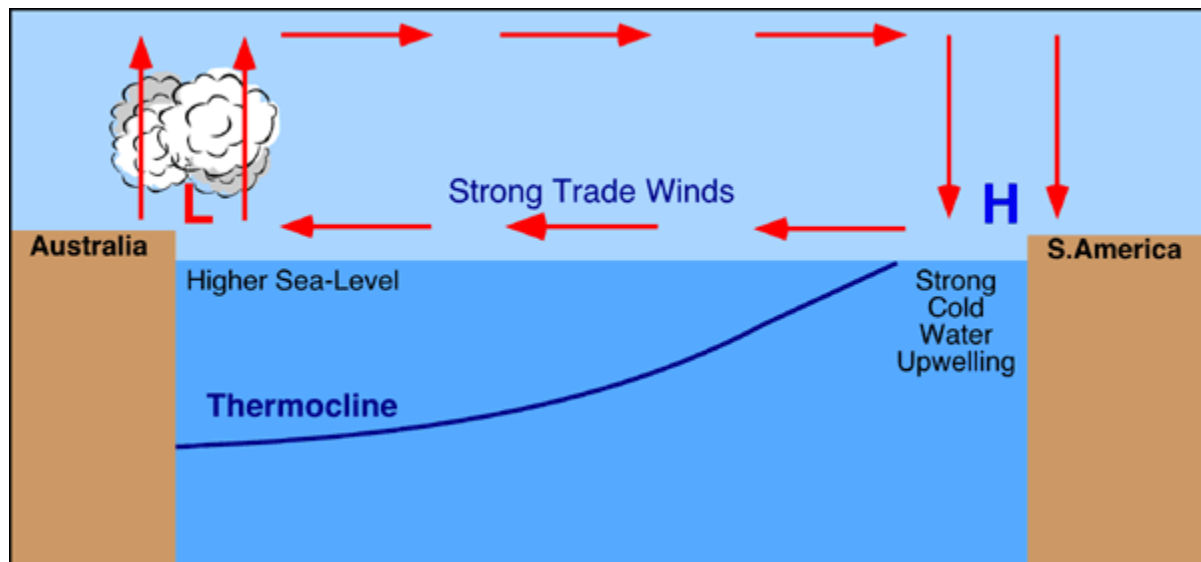
4.3 **How** much warmer than normal were the water temperatures in the North Atlantic during the El Niño event of 1911/1912? (1x1)(1)

5 degrees

4.4 If the Titanic had been launched in recent years, the disaster could have been avoided. **Why** do you think this is the case? (1x2)(2)

The Atlantic isn't experiencing an El nino event and so the sea temperatures would be colder = no ice burgs breaking off.

4.5 Study the diagrams below on monsoons and answer the questions that follow:



4.5.1 **Define** La Niña. (1x2)(2)

La Niña is a climate phenomenon characterized by unusually **cooler-than-normal sea surface temperatures** in the central and eastern tropical Pacific Ocean. It is essentially the opposite of El Niño and typically leads to changes in global weather patterns, such as increased rainfall in some regions and drought in others.

4.5.2 **Name** one major difference between La Niña and El Niño in terms of ocean temperatures. (1x2)(2)

La Niña features **cooler-than-normal** ocean surface temperatures in the central and eastern Pacific.

El Niño features **warmer-than-normal** ocean surface temperatures in the same region.

4.5.3 **Describe** two impacts of La Niña on global weather patterns. (2x2)(2)

Increased rainfall and flooding in some regions – During La Niña, areas like Southeast Asia, northern Australia, and parts of South America often experience above-normal rainfall, leading to floods (2 marks)

Drier and cooler conditions elsewhere – La Niña typically brings drier-than-normal weather to the southern United States and parts of East Africa, and can cause colder winters in some northern regions (2 marks)

4.5.4 **Explain** how La Niña can affect agriculture in southern Africa. (1x2)(2)

During a **La Niña** event, southern Africa often experiences **higher-than-normal rainfall**. This can affect agriculture in the following ways:

Positive effects: Increased rainfall can improve soil moisture and support better crop growth, potentially leading to higher yields for rain-fed crops like maize.

Negative effects: Excessive rainfall can cause flooding, waterlogging of soils, and increased pests and diseases, which can damage crops and reduce agricultural productivity.

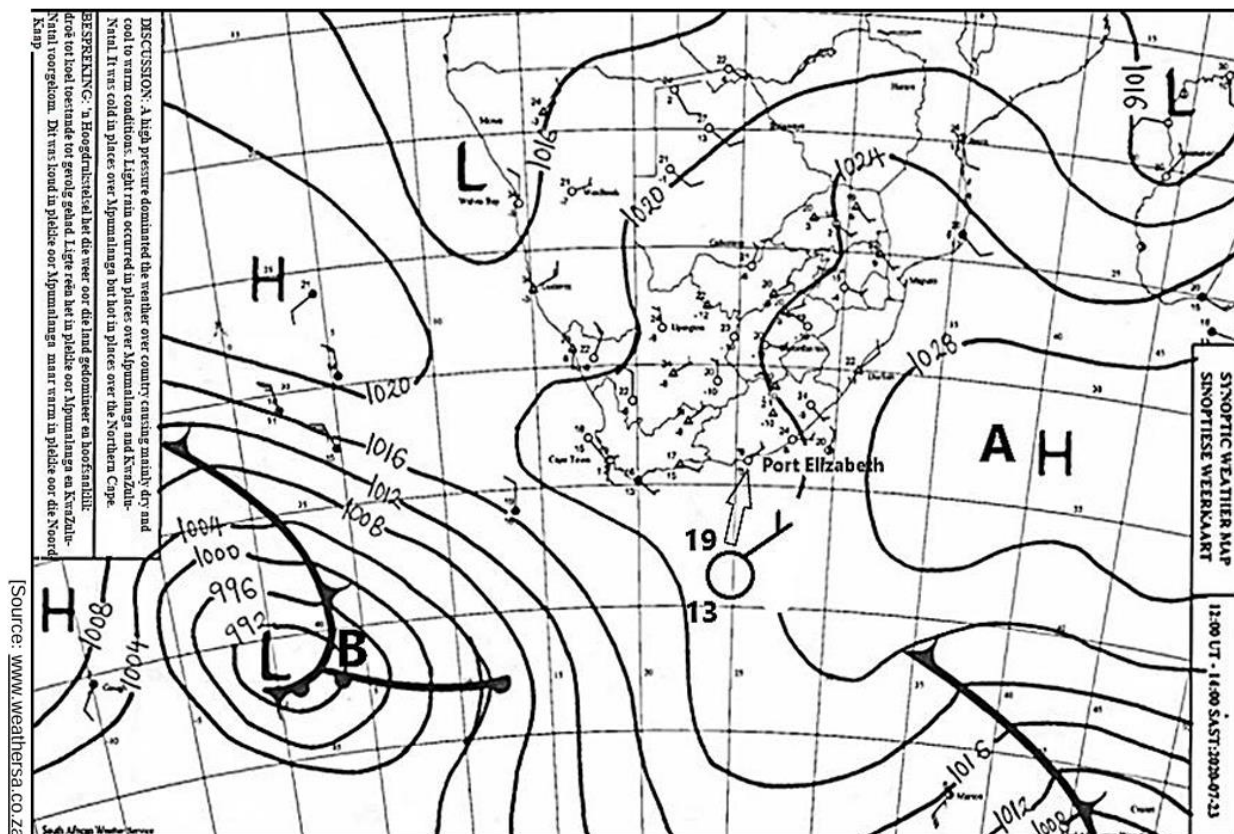
4.5.5 **Explain** why some countries may experience increased rainfall due to La Niña and explain **why**. (1x2)(2)

During a **La Niña event**, strong **easterly trade winds** push warm surface water further west across the Pacific Ocean. This allows **cold, nutrient-rich water to rise** in the eastern Pacific.

As a result, **warm ocean water pools near Southeast Asia and northern Australia**, where increased evaporation adds moisture to the air. This moisture rises and forms large rain clouds, leading to **heavier rainfall and increased storms** in those countries.

In short, the **shifting of warm water and increased convection** cause some countries to receive more rain during La Niña.

Question 5 – Synoptic weather map of South Africa [10]



5.1 **Give** evidence that the synoptic weather map represents winter conditions. (1x1)(1)

Date, Temperatures, Pressure belts.

5.2 **Determine** the isobaric interval on the synoptic weather map. (1x1)(1)

4 mb

5.3 **Give** the weather conditions of the weather station at Port Elizabeth. (5x1)(5)

19 degrees

Dew point 13 degrees

Clear skies

Wind direction NE

5 knots

5.4 **Name** the high-pressure cell at A. (1x1)(1)

South Indian High Pressure

5.5 **How** will the position of A decrease the rainfall over the eastern part of the country? (1x2)(2)

A **high-pressure system** decreases rainfall because:

The air is **descending** and warming up as it sinks.

This sinking air **prevents cloud formation** because it inhibits the rising of moist air that is needed for condensation.

Without clouds, there is **less chance of rain**, resulting in dry and stable weather.

Total for Climatology – 60 MARKS

Section B – Geomorphology [60 marks]

Question 6 – True or False [7]

State whether each of the following statements are TRUE or FALSE. Write only the question number and your answer.

- 6.1. Uniform resistance refers to rock that erodes at different rates.
- 6.2. Canyon landscapes are older than Karoo landscapes.
- 6.3. A mesa is larger than a butte.
- 6.4. Tectonic uplift and gentle, very slow flowing water results in the formation of canyons.
- 6.5. Badlands is a term used to describe a landscape with narrow gullies and sharp ridges formed in a humid climate.
- 6.6. Canyon landscapes are important tourist attractions.
- 6.7. Scarp retreat refers to the process where a slope loses width but its height remains the same.

Question 7 – Match the Column [8]

Match the following terms in column A with the correct definitions in column B. Write only the question number and the correct letter. For example, 7.9. L.

7.1. Homoclinal Ridge	A) A ridge with a gentle dip slope and steep scarp slope.
7.2. Scarp Slope	B) Erosion at the base of a slope, resulting in the collapse of the overhang above it.
7.3. Mass Wasting	C) The steep slope of a homoclinal ridge.
7.4. Undercutting	D) A ridge formed from inclined strata.
7.5. Homoclinal Shifting	E) A ridge with a steep scarp and steep dip slope.
7.6. Cuesta	F) The gentle slope of a homoclinal ridge.
7.7. Dip slope	G) The movement of a ridge towards the dip slope.
7.8. Poort	H) The movement of a ridge towards the scarp slope.
	I) Movement of eroded material downhill.
	J) Large scale dumping of mining material.

	K) The gap between two mountains.

- 6.1. False
- 6.2. False
- 6.3. True
- 6.4. True
- 6.5. False
- 6.6. True
- 6.7. True

- 7.1. D
- 7.2. C
- 7.3. I
- 7.4. B
- 7.5. G
- 7.6. A
- 7.7. F
- 7.8. K

Question 8 – Massive Igneous Rocks [15]

- 8.1 Distinguish between massive igneous rocks and stratified rocks. (2x1) (2)

Massive igneous rocks have **no layers** because they form from cooled magma, whereas **stratified rocks** are made of **distinct layers (strata)** formed by the deposition of sediments.

8.2 Draw two simple diagrams to highlight the differences between a lopolith and a laccolith.

(2x2) (4)

Laccolith and Lopolith

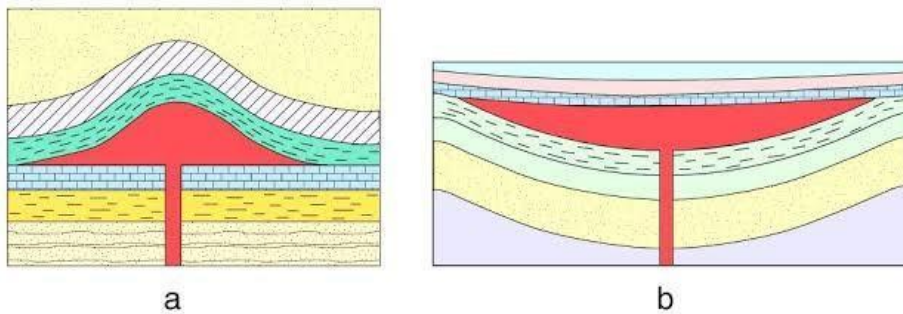


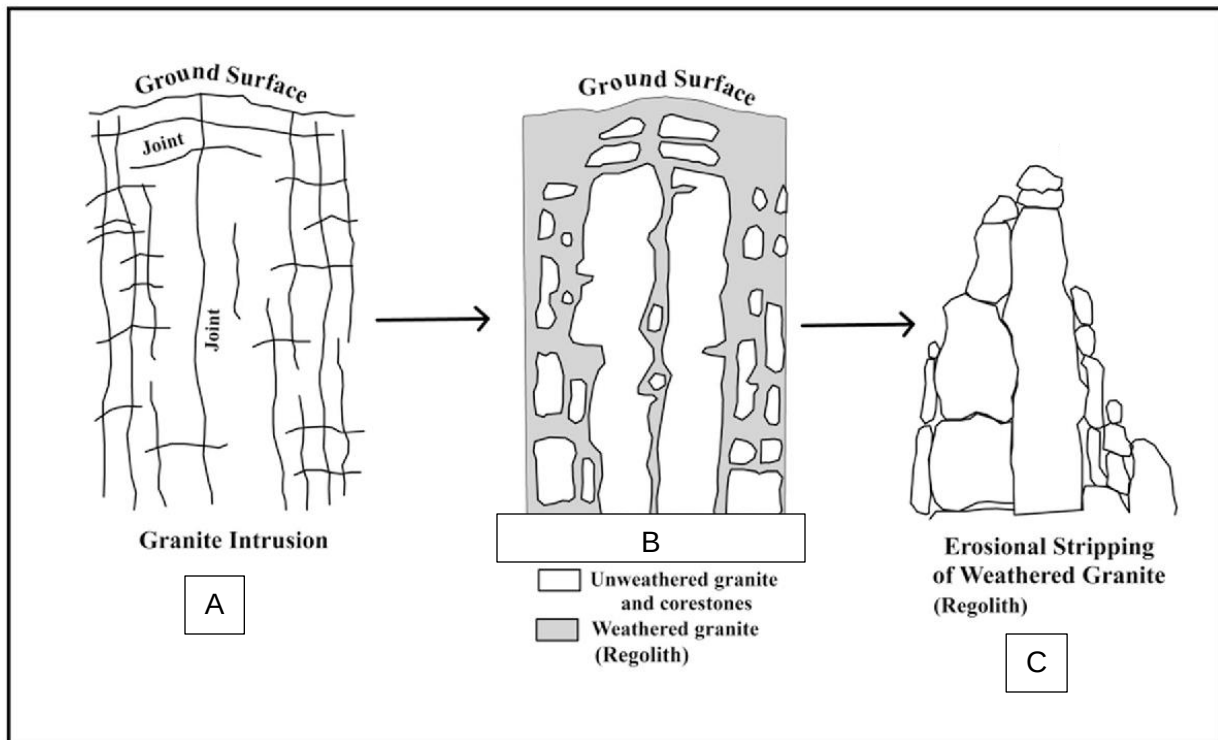
Figure 4-26. Shapes of two concordant plutons. **a.** Laccolith with flat floor and arched roof. **b.** Lopolith intruded into a structural basin. The scale is not the same for these two plutons, a lopolith is generally much larger. © John Winter and Prentice Hall.

42

2 marks = Laccolith (dome shaped)

2 marks = Lopolith (pan shaped)

Refer to the image below and answer the questions that follow:



Successive stages of tor formation

8.3 The above diagram shows the formation of a particular **landform (C)**.

The formation of **what** landform is being shown above? (1x1)(1)

A tor is being formed.

8.4 **Figure A** shows a massive igneous rock with joints in it. **Differentiate**

between contractual and off-loading joints. (2x2)(4)

Contractual joints form as a result of cooling and contraction of the igneous rock.

They occur when the hot magma solidifies and shrinks, creating cracks or fractures **(2 marks)**

Off-loading joints form as a result of pressure release.

They occur when overlying rock is removed by erosion, reducing pressure and allowing the underlying rock to expand and fracture parallel to the surface **(2 marks)**

8.5 **Describe** the process occurring at B.

(2x2)(4)

Deep chemical weathering is taking place at B.

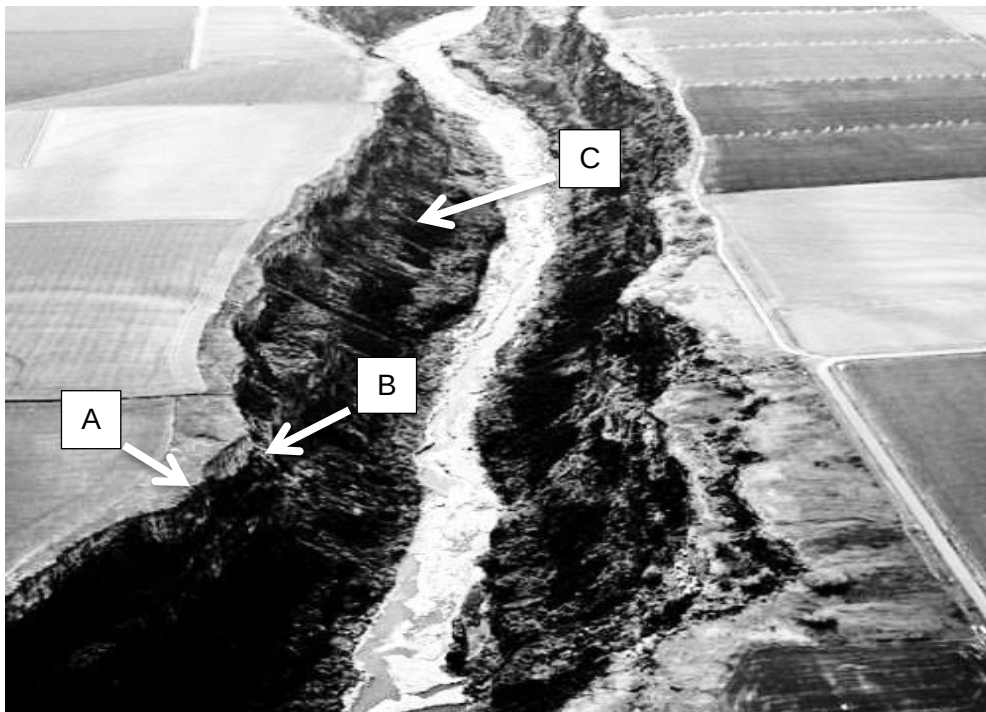
Water, often slightly acidic, percolates through joints and cracks in the granite.

Chemical reactions decompose the minerals in the granite, especially along the joints, forming weathered granite (regolith) **(2 marks)**

Rounded **corestones** of unweathered granite are left surrounded by softer, weathered material, which is eventually stripped away by erosion **(2 marks)**

Question 9 – Slopes [15]

Refer to the image below showing a river flowing in a gorge and answer the questions that follow:



- 9.1. Are the slopes on either side of this river the result of endogenic or exogenic forces? (1x1)(1)

The slopes on either side of this river are the result of **exogenic forces**.

Exogenic forces (such as weathering, erosion, and mass movement) act on the Earth's surface and shape landforms like river valleys and slopes.

- 9.2. **What** type of slopes would these be, primary erosion or secondary erosion slopes? (1x1)(1)

These would be **secondary erosion slopes**.

Secondary erosion slopes are shaped after the original land surface has been modified by agents like rivers, resulting in valley slopes due to ongoing erosion and weathering processes.

- 9.3. **Provide** labels for the slope characteristics A, B and C. (3x1)(3)

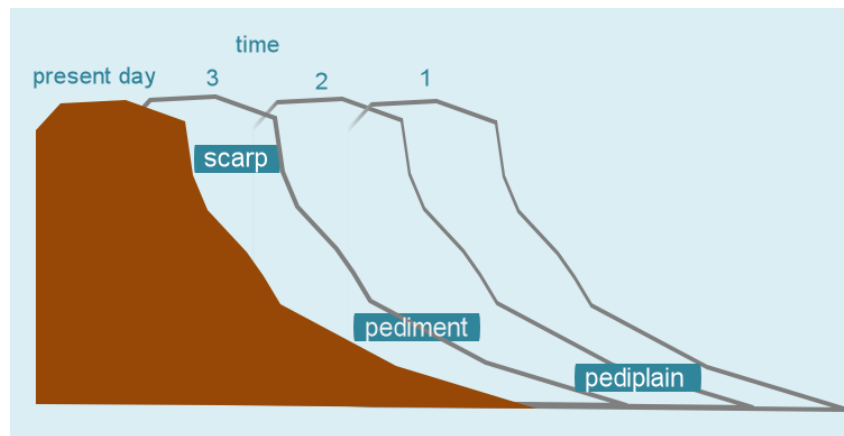
A= Plateau

B= Scarp slope

C= Talus slope

9.4. **Draw** a fully labelled diagram showing scarp retreat.

(3x2)(6)



Stage 1, 2, 3 properly drawn = 2 marks

Scarp = 1 mark

Pediment = 1 mark

Pediplain = 1 mark

Slopes eroded back = 1 mark

- 9.5. **Discuss** two ways in which the slopes in the image above can be useful to humans. (2x2)(4)

Agriculture – Gentle slopes can be terraced or used for farming (e.g. vineyards or crops), allowing people to grow food and utilize the fertile soil **(2 marks)**

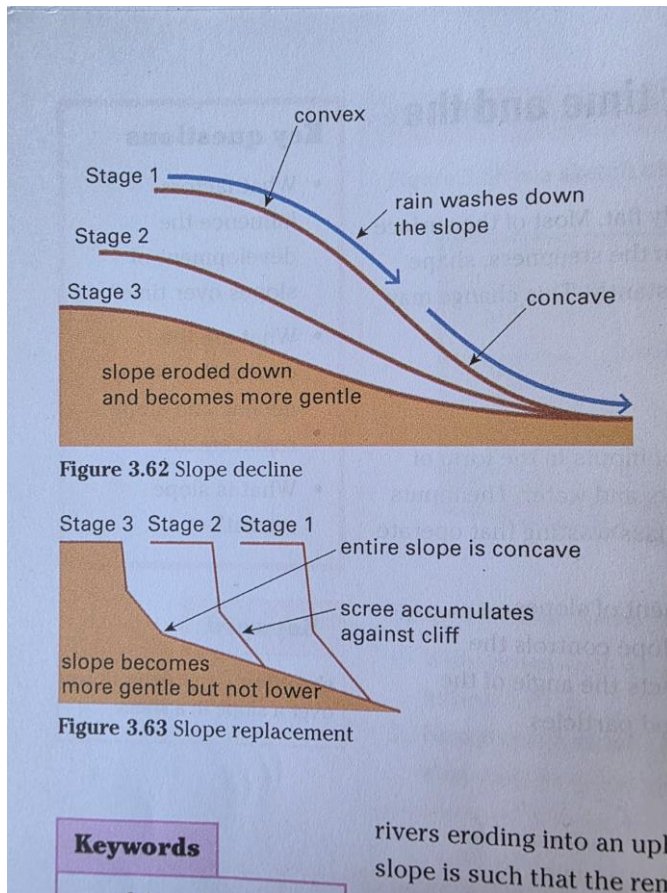
Tourism and recreation – Scenic slopes along river valleys can attract tourists for hiking, sightseeing, or adventure activities like camping, boosting the local economy **(2 marks)**

(Boys could also mention other valid points, like settlement, forestry, grazing, or hydro-electricity if relevant to the image or context.)

Question 10 – Slope theories [10]

Making use of diagrams and explanation, compare and contrast the theories of slope decline and slope replacement. Your answer should have the following:

- 10.1 **1 x diagram** showing slope decline theory. (3x1)(3)
- 1 x diagram** showing slope replacement theory. (3x1)(3)
- Brief** description of how each of these work. (2x2)(4)



Stage 1 = 1 mark

Stage 2 = 1 mark

Stage 3 = 1 mark

Dave's law for both.

Slope decline: (2 marks)

Suggests that slopes become **less steep over time**.

Weathering and erosion gradually wear the slope down, so steep slopes decrease in gradient as they age. The landscape evolves into gentler slopes as debris is removed faster than new material is supplied.

Slope replacement: (2 marks)

Suggests that slopes evolve by being **replaced by new slopes**.

Steep slopes retreat parallel to themselves as weathering and erosion cut into them, while gentler slopes build up at the base through deposition.

Question 11 – Mass Movement [5]

Study the images and description below and answer the questions that follow:



The above are images of a popular attraction in Benoni – Sandboarding. These large hills of soft sand are the result of mine dumps being repurposed for the tourism industry and have become a popular attraction for many Joburgers.

- 11.1 **Define** the term Mass Movement. (1x1)(1)
Mass movement is the downhill movement of soil, rock, and other weathered material under the influence of gravity.
- 11.2 **Discuss** two factors that may increase the rate of mass movement on these hills in Benoni. (2x2)(4)
Lack of vegetation, more people actually using the slope to sand board. Dave's law.

Total for Geomorphology – 60 MARKS

Section C – Mapwork [30 marks]

RESOURCE MATERIAL

1. 2331CC PHALABORWA and an orthophoto 2331CC PHALABORWA (North)

GENERAL INFORMATION ON PHALABORWA



Coordinates: 23°52' S ; 31°04' E

Phalaborwa is a town in South Africa. It is located near the confluence of the Ga-Selati River and the Olifants River, along the western border of the Kruger National Park in the Lowveld. Tourism and wildlife play dominant roles in the life of this town. Attractions, such as the Blyde River Canyon, the Three Rondavels, God's Window, Bourke's Luck Potholes and river cruises on the Olifants River, make Phalaborwa an important tourist destination in this province.

Question 12 – Calculations [10]

12.1 In **which** province is Phalaborwa located in? (1x1)(1)
Limpopo

12.2 **Give** the true bearing from J (C4) to H (A1) on the topographic map (use the tips of the arrows as your start and end points). (1x1)(1)
298°

12.3 **Calculate** the straight line distance of the landing strip in block C4 and B4. (1x2)(2)
(2.5cm) (0.5) = 1.25 km

12.4 **Calculate** the average gradient from trig beacon 5 in block E3 to spot height 499 in block E4. (4x1)(4)

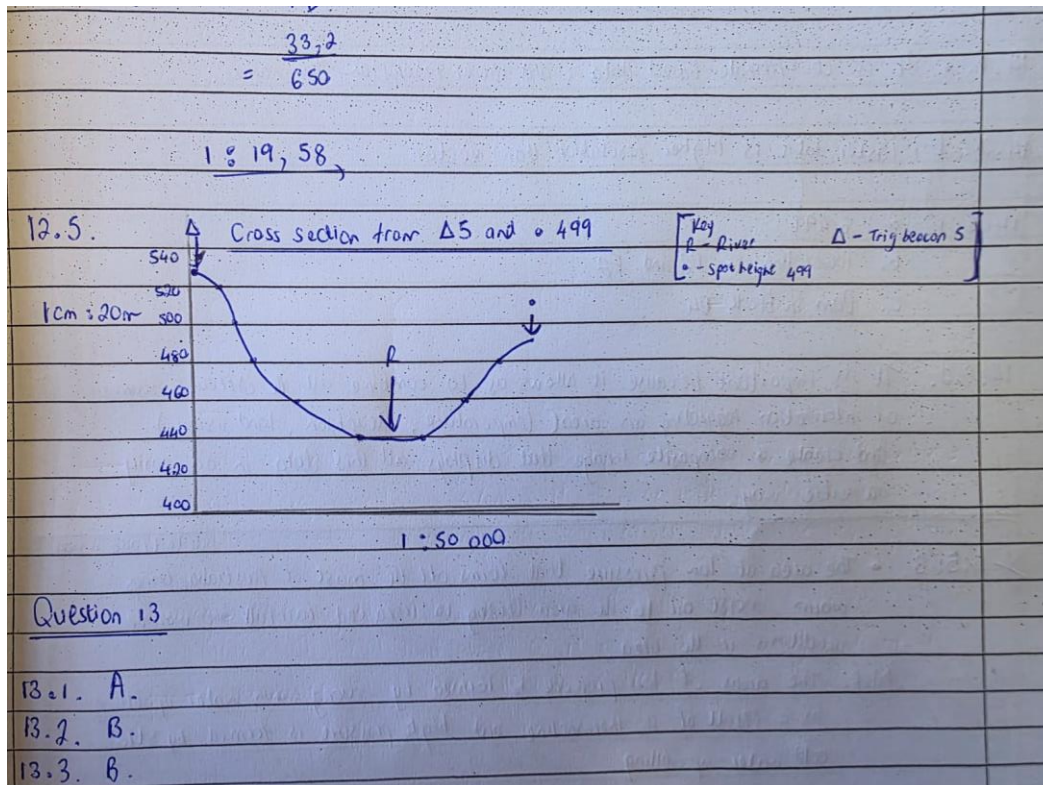
$$\text{Gradient} = \frac{VI}{HE}$$

$$HE = (1.3 \text{ cm}) (0.5) = 0,65 \text{ km} = 650\text{m} \text{ (2 marks)}$$

$$VI = 532,2 \text{ m} - 499\text{m} = 33,2\text{m} \text{ (1 mark)}$$

$$33,2/650 = 1:19,58 \text{ (1 mark)}$$

12.5 **Draw** a simple cross section between trig beacon 5 in block E3 and Spot height 499 in block E4. (2x1)(2)



Trend = 1 mark

Heading, scale and Key = 1 mark

Question 13 – Interpretation [12]

13. Select the correct answer from the options provided with each question below:

13.1 The river in block D1 is flowing in a ... direction.

A Northerly.

B Southerly

C Easterly

D Westerly

(1x1)(1)

13.2 The highest point in block D2 is ...

A Spot height 391

B Trig beacon 12

C The bench mark.

D Spot height 450.

(1 x 1) (1)

13.3. The dominant street pattern in block D5 on the topographic map is ...

A unplanned.

B planned irregular.

C radial.

D grid iron.

(1x1)(1)

13.4. **Provide** ONE benefit of this pattern answered above.

(1x1)(1)

Traffic alleviation

13.5. **What** economic activity is likely taking place in block C4 and B4?

(1x1)(1)

Flying, private air strip

13.6. **Identify** the predominant land usage in block C1. (1x1)(1)
Farming/ Agriculture

13.7. **Give** TWO pieces of evidence from the map to motivate the location of the sewerage works in block D2. (2x1)(2)
Located far away from a residential area.
It is near a river.

13.8. Urbanisation is increasing in the Phalaborwa region.
Discuss why the urbanisation of Phalaborwa can take place in any direction. You must **provide** evidence from the map. (2x2)(4)

Gradient: The contour lines are spread far apart meaning that it is Relatively flat grown which is good for building.

Open/Free land: There is actual space to expand into.

Non-perennial rivers: Fewer water obstacles means expansion is Easier. ANY REASONABLE ANSWER

Question 14 – GIS [8]

Refer to the images below illustrating vector and raster data and answer the questions that follow:



14.1 **Define** the term vector data. (1x1)(1)

Vector data is a type of geographic data that uses points, lines, and polygons to represent real-world features and their precise locations on the Earth's surface.

14.2 **Which** image, A or B, is raster data?

Give a reason for your answer. (1+1)(2)

A, it is cell based and less clear.

14.3 Refer to block D3 on the topographical map.

14.3.1 Identify the following in block D3:

(a) A point feature (1x1)(1)

(b) A line feature (1x1)(1)

(c) An area/polygon feature (1x1)(1)

Any point, Line or polygon.

14.3.2 **Explain** why layering is important in GIS. (1x2)(2)

Any reasonable answer.

Total for Mapwork – 30 MARKS

Grand Total – 150 MARKS